

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Header	Information
Pentester Name	Muhammad Zain Ul Abiddin
Program/Batch	B083A - Networkwalks
Date	13 September 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission Secured?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting

Header	Information
	Phase 2: Scanning & Network Discovery Phase 3–5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorized access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing training program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities.

Tool	Purpose
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs, and MAC addresses.
Windows CMD	Local IP and MAC address identification.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon. Each tool was used to collect a different type of information about the target.

1. **WHOIS:** Used whois networkwalks.com to obtain publicly available domain registration information and identify the domain's name servers (HostGator).
2. **WhatWeb:** Executed whatweb networkwalks.com to identify technologies used by the website. Results identified WordPress 7.0.4, WP Download Manager 3.3.58, Apache web server, and Bootstrap.
3. **Nslookup:** Ran nslookup networkwalks.com to resolve the domain name to its IPv4 address (192.232.216.135).

4. **Curl:** Used curl -I https://networkwalks.com to inspect HTTP response headers. Exposed the WordPress REST API endpoint /wp-json/ and caching details.
5. **Wafw00f:** Executed wafw00f networkwalks.com to test for firewall protection. Result confirmed ModSecurity (SpiderLabs) WAF.
6. **DNSRecon:** Ran dnsrecon -d networkwalks.com to enumerate DNS records including name servers, MX mail servers, SPF policies, and SRV service records.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap on my Windows host to perform network discovery on my local LAN.

1. **IP Identification:** Used Windows ipconfig command to identify my local IP address (192.168.86.35) and LAN subnet (192.168.86.0/24).
2. **Ping Scan:** Entered 192.168.86.0/24 into Zenmap and selected **Ping Scan** (nmap -sn) to identify active hosts.
3. **Host & MAC Discovery:** Extracted live host IP addresses and MAC addresses across the subnet.
4. **Topology Export:** Opened the **Topology** section in Zenmap, enabled the legend, and exported the network topology in PDF format.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed version details to search public exploit databases	● Medium

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
2	Server IP address identifiable	Nslookup resolved domain to 192.232.216.135	Provides exact network location of target server	● Low
3	HTTP technical information exposed	Curl exposed response headers and /wp-json/ REST API	Assists in technology fingerprinting and hidden route enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Exposes defense mechanisms, allowing attackers to plan bypass techniques	● Low
5	DNS infrastructure information exposed	DNSRecon identified mail, SPF, and service records	Reveals external mail servers and host organization structure	● Medium
6	Live hosts visible on local network	Zenmap identified active hosts across local subnet	Unauthorized or rogue devices could exist undetected on LAN	● Medium

Risk level key: ● Critical | ● Medium | ● Low

Note: The risks above are observations from footprinting and scanning, not verified exploits. No unauthorized exploitation was performed.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review Exposed Technology Information:** Minimize public software version disclosure in HTTP headers and HTML metas.
2. **Keep Software Updated:** Frequently patch WordPress core, plugins, and web servers to mitigate public vulnerabilities.
3. **Harden HTTP Headers:** Configure web servers to strip banners like Server: Apache and hide internal API endpoints where possible.
4. **Audit DNS Records:** Periodically review public DNS zone files to ensure unnecessary subdomains or internal IP leaks are removed.
5. **Tune WAF Rules:** Ensure ModSecurity rules remain updated to block evolving automated scanning and bypass attempts.
6. **Conduct Regular Subnet Scans:** Perform periodic internal network audits to detect unauthorized devices.
7. **Document Network Topology:** Maintain updated network maps and device inventories for security monitoring.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking program at Networkwalks, I completed practical activities covering footprinting, reconnaissance, and network scanning.

The exercises demonstrated that passive information gathering is an essential first phase of ethical hacking. Security professionals can uncover significant infrastructure details without firing a single attack payload. Proper documentation and clear risk reporting are vital for remediating leaks before malicious actors can exploit them.